Prenatal Genetic Testing

(Screening and diagnosis of genetic diseases)

Genetic screening

- The process of identifying persons at risk of producing children with genetic defects and of assessing the genetic state of early embryos and neonates is called genetic screening.

Types of prenatal test:

Today many prenatal tests are available to detect chromosomal and gene mutation. Karyotyping of fetal or embryonic cells can indicate whether a developing child has Down syndrome or one of the other chromosomal abnormalities. Biochemical tests are available for dozens of enzymes such as hexosaminidase-A the missing enzyme in persons with Tay-sachs disease

- Some prenatal tests are screening tests, while others are diagnostic.
- A screening test gives information about the possibility that a problem exists. For example, an α-fetoprotein (AFP) screening test can show that there is a 1% risk of Neural tube defects.

- A diagnostic test can tell that a specific problem is present (e.g., amniocentesis) is offered to confirm or exclude the presence of trisomy 21 and other chromosome abnormalities.

Diagnostic prenatal testing can be done by invasive or non-invasive and less invasive methods.

- An invasive method involves probes or needles being inserted into the amniotic sac, placenta, or umbilical cord e.g. amniocentesis, chorionic villus sampling, and fetal blood sampling.

-Non-invasive techniques include examinations of the woman's womb through ultrasonography and maternal serum screening (i.e. Alpha-fetoprotien , fetal cell and cell-free fetal DNA in maternal blood).

Reasons for prenatal diagnosis

1-To enable timely medical or surgical treatment of a condition before or after birth.

2-To give the parents the chance to abort a fetus with the diagnosed diseased condition.

3-To give parents the chance to prepare psychologically, socially, financially, and medically for a baby with a health problem or disability, or for the likelihood of a stillbirth

Who should consider having a prenatal gen. diagnostic test?

1- Mothers older than 34 years old.

2- Mothers whose ethnic background has been associated with certain types of genetic disorders such as Tay-Sachs disease (Ashkenazi Jewish), sickle cell anemia (African American background) or cystic fibrosis (Caucasian background).

3- Mothers who have a family history or who had a previous child with a genetic condition.

4-Mothers who have experienced multiple miscarriages.

5- Mothers who had a prenatal screening test that showed they were at increased risk of having a baby with a chromosomal abnormality such as DS, trisomy 18 or spina bifida.

6- Close related couples(first cousin)

7-Women exposed to adverse environmental factors.

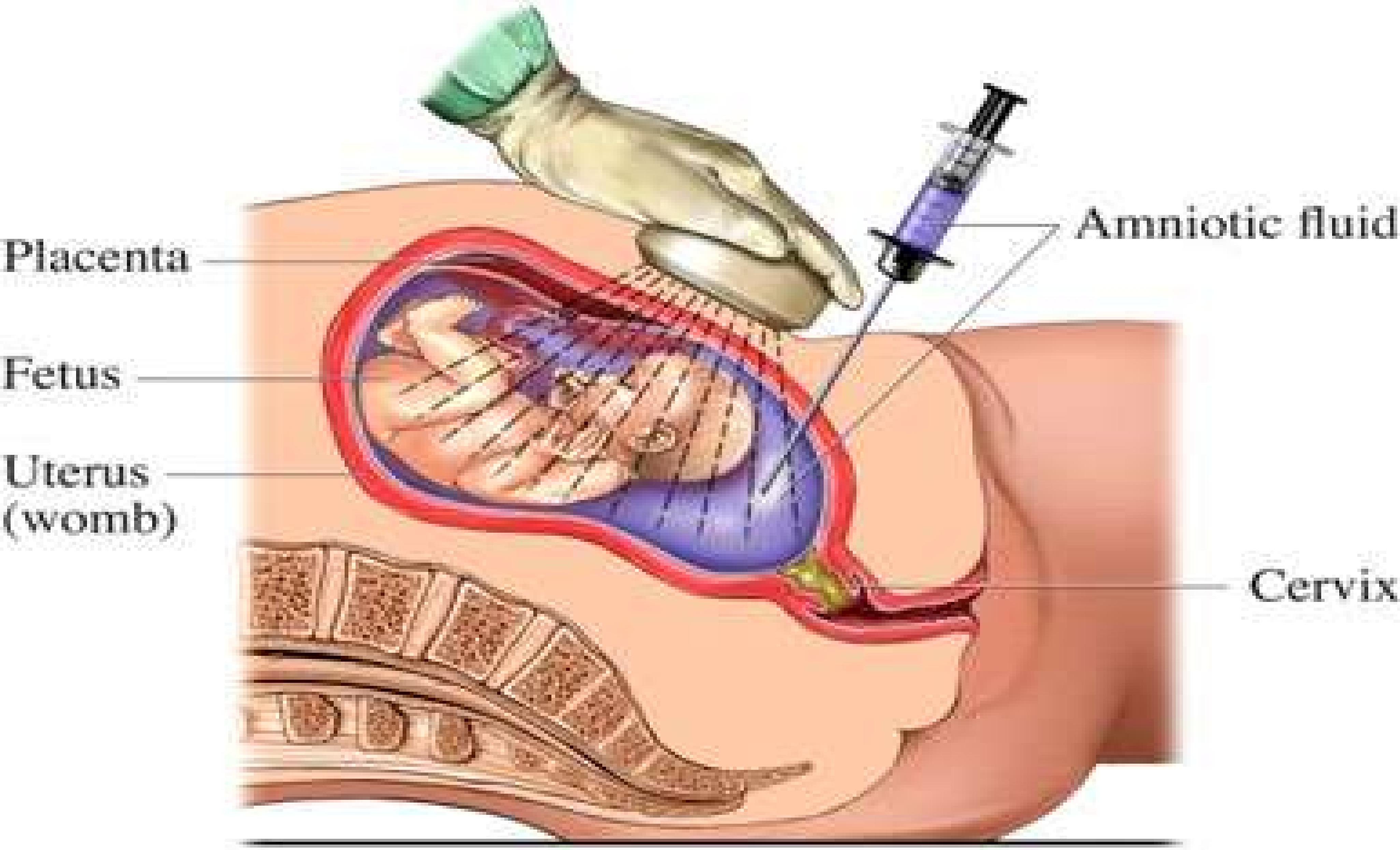
Amniocentesis

-In the 4th months of pregnancy, a small sample of amniotic fluid is removed by means of a sterile needle.

- A needle is inserted through the abdominal wall in to the amniotic sac after ultrasound localizes the placenta and determines the position of fetus.

-A bout 20-30 ml of amniotic fluid is with drown; this fluid contains living cells (**amniocyte**) shed by the fetus.

-These cells can be grown in tissue culture and multiplied in adequate numbers for testing purpose, (Karyotyping, biochemical tests and DNA analyses).

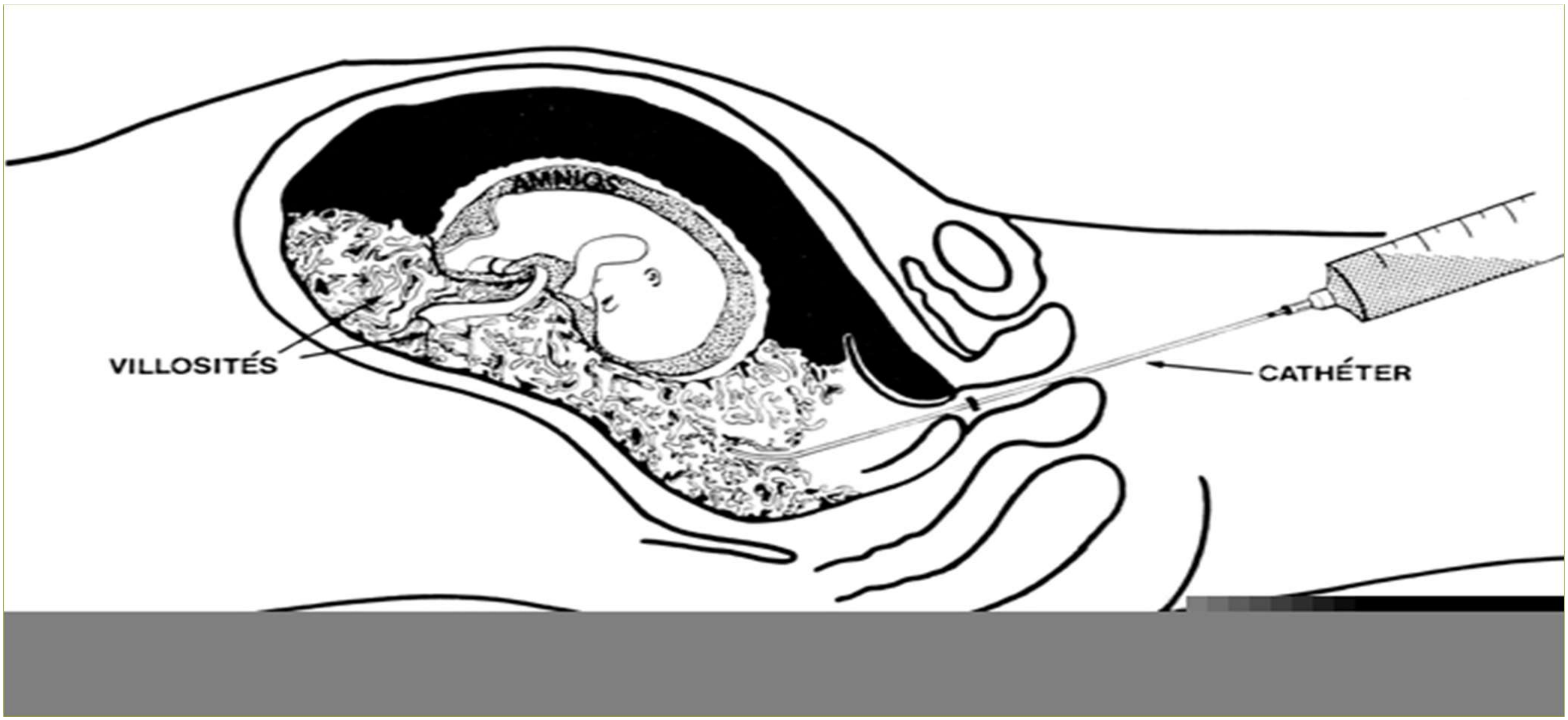

Placenta
Fetus
Uterus
(womb)
Amniotic fluid
Cervix

Chorionic villus sampling

-CVS is performed by aspirating fetal trophoblastic tissue (**chorionic villi**) by either a trans cervical or trans abdominal approach.

- It is usually performed at 10 weeks of pregnancy. The chorionic villi contain cells with same chromosomal and genetic make up as fetus and therefore these cells can be tested for possible fetal abnormalities.

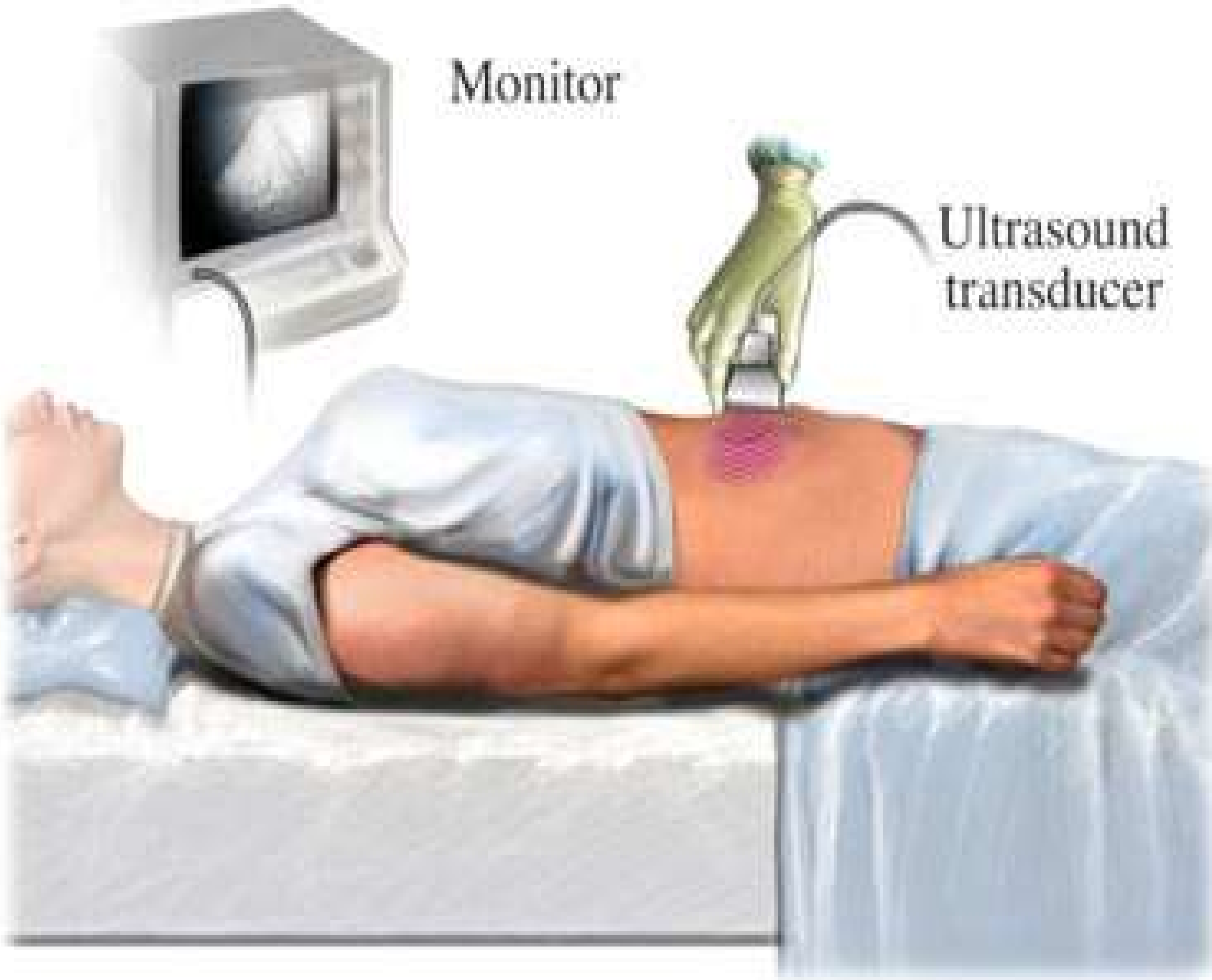
-The advantage of CVS is that testing can be done much earlier and an abortion if desired can be done during the 1st trimester when it is considered medically safer for the mother.



Ultrasound

-Ultrasonography is the most common technique now used for fetal visualization.

-It is used to confirm a pregnancy and to test for a specific congenital malformations e.g: a short limb skeletal dysplasia(dwarf), anecephaly, spinabifida and poor fetal growth. Sometimes it can be used to evaluate the uncertain gestational age, amniotic fluid volume and determine the baby’s sex.


Monitor
Ultrasound
transducer

Maternal blood sampling

-Some genetic conditions can be detected by mother serum screening ,for instance , we can check levels of important markers produced by the fetus and placenta (alpha-fetoprotein , estriol ,β-hCG , inhibin-A,) in the woman's serum.

-The level of alpha-fetoprotein is significantly higher than normal when fetus has a neural-tube defects or may be decreased in the case of Down syndrome.

Fetal cell sorting

-During pregnancy ,a few fetal cells are released into the mother’s circulatory system.

- Recent advances have made it possible to separate fetal cells from a maternal blood sample (automated cell-sorting machines) fetal cells can be detected and separated from maternal blood .

-The fetal cells obtained can be cultured for chromosome analysis or DNA molecular testing